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# Early Devonian paleomagnetic data from the Lower Devonian Karheen Formation suggest Laurentia-Baltica connection for the Alexander terrane

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## ABSTRACT

**Paleomagnetic data from 29 sites within the Lower Devonian Karheen Formation pass fold and reversal tests, and define two polarity zones that are correlative between two separate stratigraphic sections. We interpret this magnetization to be Early Devonian and thus propose a paleopole for the Alexander terrane that places the terrane at about 14 (north or south). Due to uncertainties of the Early Devonian parts of the North American and Gondwana apparent polar wander paths, several interpretations of the Early Devonian position of the Alexander terrane relative to either North America (Laurentia) or Gondwana are permissible. We use a combined data set of paleomagnetic, detrital zircon, and paleontological observations to narrow the paleogeographic options. Eastern Australia and the Scandinavian part of Baltica lie within the paleolatitude bands, and both are possible sources of Precambrian detrital zircon grains found in the Karheen Formation; however, paleontological data from the underlying Heceta Formation are only consistent with a position near northern Laurentia-Baltica. Therefore, the combined data sets indicate an Early Devonian position near Baltica.**

## INTRODUCTION

We sampled Paleozoic strata in the Prince of Wales Island area of southeastern Alaska (Fig. 1) to determine if the paleomagnetism and detrital zircon characteristics of these rocks define the paleolatitudinal motion history of the Alexander terrane. Pole positions from a previous study of the Prince of Wales Island area of the Alexander terrane (Van der Voo et al., 1980) were interpreted as evidence that the Paleozoic paleolatitudes for the Alexander terrane were lower than predicted for southeastern Alaska from the Paleozoic apparent polar wander (APW) path of North America. However, the best determined of these pole positions were based on five or fewer sites and, with revised Paleozoic North American reference poles for comparison, a paleoposition tied to North America is not entirely consistent with the data obtained. Furthermore, recent studies suggest that the characteristic remanent magnetizations (ChRMs) of at least two of the formations (the Devonian Wadleigh and Mississippian Peratrovich formations) studied by Van der Voo et al. (1980) are secondary magnetizations (Butler et al., 1993). New paleomagnetic data obtained from the Lower Devonian red beds of the Karheen Formation provide a more definitive Early Devonian paleolatitude of 14° (N or S).

## GEOLOGIC SETTING

The Alexander terrane is a tectonic fragment that underlies much of southeastern Alaska, the Saint Elias Mountains of British Columbia and Yukon, and coastal British Columbia (Berg et al., 1972). These strata are depositionally overlain along the eastern margin of the terrane by Upper Jurassic–Lower Cretaceous strata of the Gravina belt, which in turn are bound to the east by mid-Cretaceous thrust faults that record structural accretion of the terrane.

The Karheen Formation in the northern Prince of Wales Island region is composed of conglomerate, sandstone, mudstone, and subordinate limestone (Eberlein et al., 1983). The age of the Karheen Formation is determined paleontologically. The underlying Heceta Formation contains Early to Late Silurian (Llandovery-Ludlow) body and trace fossils (Soja, 1991). The base of the overlying Wadleigh Formation and the upper part of the Karheen Formation contain late Early Devonian conodonts, and the middle Karheen strata contain early Pragian (middle Early Devonian) conodonts (Savage, 1977, 1981; Savage and Gehrels, 1984; Soja, 1988). These associations indicate that the Karheen Formation is middle Early Devonian in age.

We studied two continuous stratigraphic sections of the Karheen Formation located on the northwestern part of Prince of Wales Island and adjacent Tuxekan and Heceta islands (Fig. 1). The western section has an apparent thickness of >1000 m. Bedding dips range from nearly flat at the axis of the syncline in the eastern section to nearly vertical (80°) in the western section.

The rocks vary from coarse-grained to very fine grained arkosic sandstones to fine-grained mudstones. An opaque, red-brown matrix contributes to the red-bed appearance of this unit. Reflected-light microscopy has revealed an abundance of rounded

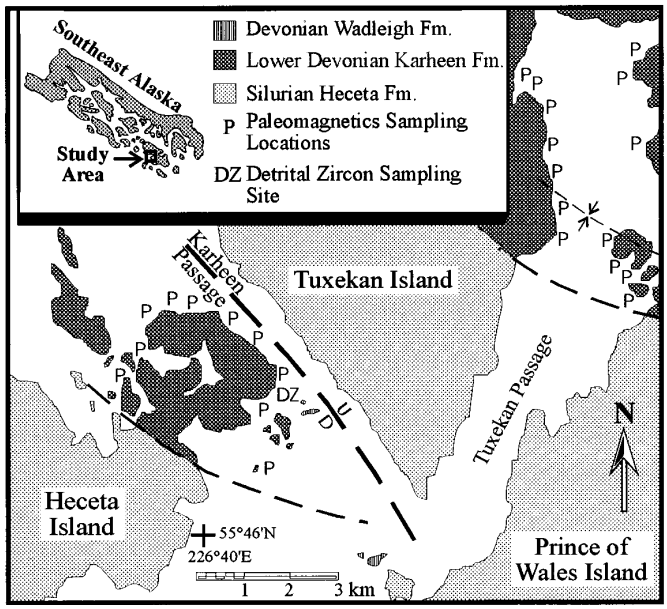



**Figure 1. Location map showing study area and names and ages of formations sampled for paleomagnetic and detrital zircon analysis. U and D, up and down, indicate motion on proposed fault. Geology from Eberlein et al. (1983).**

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TABLE 1A. KARHEEN FORMATION, SITE-MEAN DIRECTIONS (CORRECTED FOR BEDDING TILT)

| Site | $N/N_o$ | DipAz/Dip °E / ° | D (°) | I (°) | k | $\alpha_{95}$ (°) | PLAT (°N) | PLON (°E) |
|---|---|---|---|---|---|---|---|---|
| CR030 | 6 / 6 | 85 / 61 | 64.3 | -51.5 | 176 | 5.1 | -13.5 | 355.0 |
| KR010 | 6 / 6 | 82 / 54 | 63.0 | -41.5 | 130 | 5.9 | -5.8 | 351.7 |
| KR011 | 5 / 6 | 79 / 47 | 77.3 | -23.0 | 30 | 14.1 | -2.9 | 333.8 |
| KR012 | 4 / 6 | 80 / 64 | 46.7 | -41.3 | 66 | 11.4 | 1.1 | 4.9 |
| KR013 | 5 / 6 | 45 / 85 | 71.1 | -55.1 | 81 | 8.6 | -19.5 | 352.0 |
| KR015 | 6 / 6 | 81 / 59 | 52.0 | -26.8 | 96 | 6.9 | 7.7 | 356.3 |
| KR016 | 6 / 6 | 57 / 74 | 52.0 | -13.9 | 393 | 3.4 | 14.0 | 353.0 |
| KR020 | 6 / 6 | 5 / 26 | 227.4 | 29.7 | 66 | 8.3 | -8.0 | 181.2 |
| KR021 | 5 / 6 | 9 / 23 | 225.4 | 19.8 | 225 | 5.1 | -14.0 | 180.6 |
| KR040 | 5 / 6 | 6 / 26 | 227.1 | 34.2 | 27 | 14.9 | -5.5 | 182.6 |
| KR041 | 4 / 6 | 6 / 26 | 231.4 | 39.0 | 15 | 24.6 | -1.0 | 180.4 |
| KR042 | 5 / 6 | 6 / 26 | 216.5 | 40.7 | 40 | 12.2 | -5.1 | 193.6 |
| KR043 | 6 / 6 | 6 / 26 | 216.7 | 32.0 | 31 | 12.2 | -10.6 | 191.3 |
| KR044 | 5 / 6 | 3 / 23 | 214.4 | 18.8 | 46 | 11.4 | -18.6 | 190.9 |
| KR045 | 5 / 6 | 10 / 24 | 217.9 | 36.7 | 117 | 7.1 | -7.3 | 191.3 |
| KR058 | 6 / 6 | 93 / 35 | 44.9 | -19.3 | 29 | 12.8 | 14.4 | 0.9 |
| KR064 | 4 / 6 | 133 / 11 | 32.8 | -5.9 | 118 | 8.5 | 25.4 | 10.0 |
| KR066 | 4 / 6 | 75 / 76 | 58.5 | -30.0 | 114 | 8.7 | 3.0 | 351.6 |
| KR067 | 4 / 6 | 75 / 76 | 66.3 | -46.2 | 16 | 23.8 | -10.5 | 351.0 |
| KR069 | 4 / 6 | 82 / 58 | 60.0 | -24.1 | 78 | 10.4 | 5.4 | 348.6 |
| KR070 | 4 / 6 | 80 / 58 | 73.7 | -29.3 | 13 | 26.1 | -4.1 | 338.8 |
| KR072 | 6 / 6 | 88 / 60 | 45.8 | 19.9 | 14 | 18.8 | 32.2 | 350.2 |
| KR075 | 6 / 6 | 78 / 78 | 68.6 | -27.1 | 23 | 14.3 | -0.3 | 342.3 |
| KR076 | 6 / 6 | 80 / 86 | 55.4 | -31.9 | 197 | 4.8 | 3.4 | 354.7 |
| KR080 | 6 / 6 | 48 / 42 | 49.9 | 10.1 | 72 | 8.0 | 25.7 | 348.9 |
| KR084 | 7 / 7 | 97 / 28 | 73.8 | 8.4 | 8 | 23.2 | 12.5 | 327.8 |
| KR085 | 6 / 6 | 66 / 48 | 230.3 | 19.7 | 19 | 18.8 | -12.0 | 175.9 |
| KR087 | 6 / 6 | 33 / 26 | 223.0 | 4.7 | 202 | 4.7 | -22.2 | 179.3 |
| KR088 | 6 / 6 | 55 / 44 | 56.4 | -17.4 | 10 | 22.5 | 10.4 | 349.9 |

TABLE 1B. FORMATION MEAN DIRECTIONS AND PALEOMAGNETIC POLE

| N | D (°) | I (°) | k | $\alpha_{95}$ (°) | PLAT (°N) | PLON (°E) | K | $A_{95}$ (°) |
|---|---|---|---|---|---|---|---|---|
| Geographic coordinates (uncorrected for tilt) | | | | | | | | |
| 29 | 230.0 | -19.8 | 9 | 9.4 | | | | |
| Stratigraphic coordinates (corrected for tilt) | | | | | | | | |
| 19p | 58.0 | -24.1 | 13 | 9.8 | | | | |
| 10p | 223.0 | 27.7 | 41 | 7.6 | | | | |
| 29 | 232.7 | 25.6 | 15 | 7.1 | -7.1 | 175.4 | 26 | 5.4 |

Note: N-number of specimens or sites used to determine site-mean directions, formation-mean direction, polarity-mean direction, and/or paleomagnetic pole, (p = number of sites of one polarity); $N_o$-number of specimens thermally demagnetized; DipAz-azimuth of down-dip direction; D-mean declination; I-mean inclination; k estimate of Fisher precision parameter; $\alpha_{95}$-radius of the cone of 95% confidence about the mean direction; PLAT-latitude of VGP or paleomagnetic pole; PLON-longitude of virtual geomagnetic pole (VGP) or paleomagnetic pole; K-estimate of Fisher precision parameter for paleomagnetic pole; $A_{95}$-radius of the cone of 95% confidence about the paleomagnetic pole position. Paleomagnetic pole calculated from site VGPs.

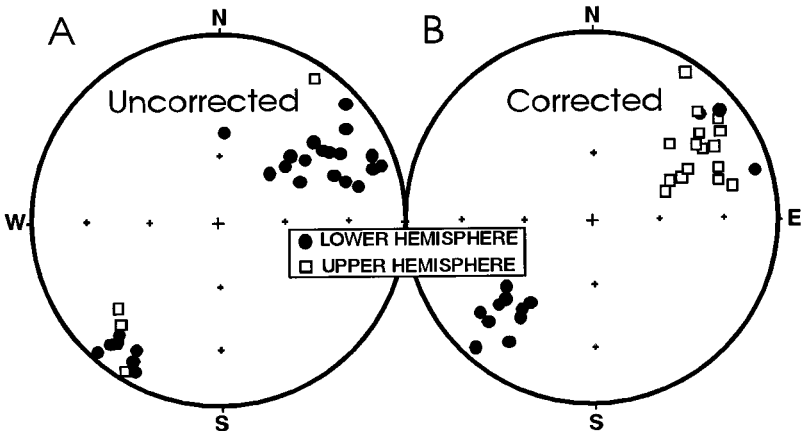



**Figure 2. Equal-area projections of site-mean directions before (A) and after (B) correction for tilt of strata.**

opaque grains with hematite cores. A few grains display relict magnetite-ilmenite intergrowths, indicating that magnetite has been replaced by hematite (martite). No evidence of high-temperature exsolution of ilmenite was observed in the hematite grains; therefore we interpret the hematite as having formed in situ shortly after deposition.

## PALEOMAGNETIC DATA AND ANALYSIS

We collected 6 to 10 oriented cores from each of 68 horizons (= site) within the two continuous stratigraphic sections described above. Remanent magnetizations were measured with a two-axis cryogenic magnetometer, and the hematite-bearing samples were thermally demagnetized by using shielded furnaces. Secondary components were removed by demagnetization up to 550 °C and the ChRM is isolated in many specimens between 550 and 680 °C. For several sites, demagnetization of pilot samples above 500 °C produced only erratic magnetic behavior. We used isothermal remanent magnetization (IRM) experiments to study the nature of the carrier of the erratic magnetizations. This analysis showed that in some specimens magnetite generated during thermal demagnetization results in acquisition of a magnetization in magnetite that masks any ChRM carried by the original hematite.

Because of these complications, further efforts were concentrated on retrieving data from sites with a well-defined and well-grouped ChRM. To define the ChRM accurately, at least five thermal demagnetization steps were included between 600 and 680 °C. Principal component analysis (Kirschvink, 1980) was used to determine the ChRM direction of each specimen, and these ChRMs were used to calculate the site-mean directions (Table 1A, Fig. 2). Site-mean directions with $\alpha_{95} > 30°$ and two site-mean directions that pass the above criteria but are more than two angular standard deviations from the mean of all site-means are excluded from Table 1A and Figure 2. We suspect both of the anomalous directions are due to local, undetected rotations of the outcrops.

The age of the Karheen ChRM, and thus the age of the paleomagnetic pole, is interpreted to be contemporaneous with the Early Devonian age of deposition. This is based on the following observations. (1) The site-mean directions pass a reversal test. The null hypothesis that the two samples are drawn from populations with means that are 180° apart is accepted at a >95% confidence level. These directions could be resolved to better than 20°, but not better than 10°, so the classification of the reversal test is C (McFadden and McElhinny, 1990); the crit-

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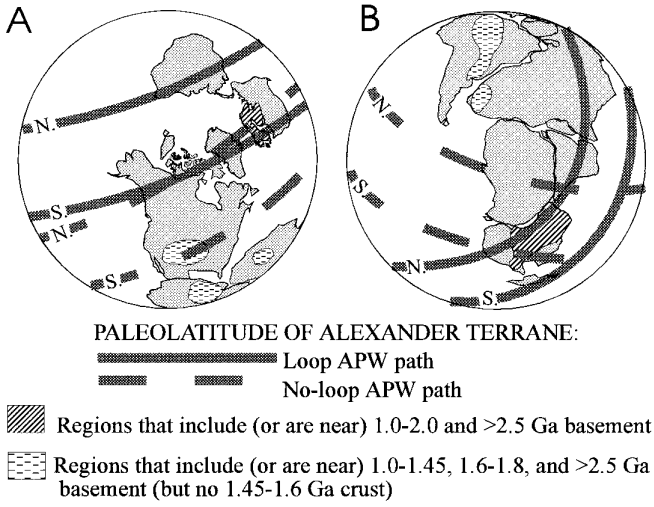



**Figure 3. Early Devonian paleolatitudes determined from Karheen Formation paleopole relative to Laurentia and Gondwana. Paleogeographies of North America, Siberia, Baltica, and Gondwana are based on reference paleopoles (Van der Voo, 1993). A: Two pairs of possible Karheen paleolatitudes (14° ± 5° N or S) relative to North America, Siberia, and Baltica. Loop paleolatitudes are determined assuming that west longitude loop exists in Early Devonian part of North American apparent polar wander (APW) path. No-loop paleolatitudes are determined assuming that no loop exists in North American APW path. B: Two pairs of possible Karheen paleolatitudes (14° ± 5° N or S) relative to Gondwana. Loop paleolatitudes are determined assuming that loop exists in Early Devonian part of Gondwana APW path. No-loop paleolatitudes are determined assuming that no loop exists in Gondwana APW path. Precambrian basement provinces were compiled from Hoffman (1989) and Goodwin (1991).**

ical angle is 15°. In addition, mixed polarities are typical of rocks considered to be reliable recorders of the Early Devonian magnetic field (Mac Niocaill and Smethurst, 1994). (2) The polarity of strata is correlative between the two sections sampled. At the base of each section, the ChRM is inclined down to the southwest; higher in each section, the ChRM is inclined up to the northeast. (3) There is a significant (>95%) correlation between the dip directions and the in situ (uncorrected) ChRM directions. Maximum grouping is achieved at 92% unfolding, but the grouping of directions is not significantly different than at 100% unfolding. At both 92% and 100% unfolding there is no significant (>95%) correlation between the dip directions and the ChRM directions, indicating that the ChRM was acquired prior to folding (McFadden, 1990). The age of the folding is not well known, so the positive fold test only indicates that the ChRM is at least pre–mid-Cretaceous.

## INTERPRETATION AND TERRANE HISTORY IMPLICATIONS

Our interpretation is that the Karheen ChRM was acquired shortly after deposition, and we propose a middle Early Devonian paleopole for the Alexander terrane near 7.1°S, 175.4°E (Table 1B). This paleopole places the Alexander terrane at 14° ± 5° (north or south) latitude during the Early Devonian.

Detrital zircons collected from the Lower Devonian Karheen Formation at one location (Fig. 1) include grains yielding Ordovician-Silurian, 1.0–1.2, 1.35–1.39, 1.48–1.53, 1.62–1.68, 1.73–1.77, 1.8–2.0, and 2.5–3.0 Ga dates. The Proterozoic and Archean grains suggest that the Alexander terrane was close to continental crust (of these ages) during the Early Devonian. Unfortunately, determining a unique position of the Alexander relative to North America (Laurentia) or Gondwana is complicated by the uncertainties of global paleogeography for Silurian-Devonian time and by the hemisphere ambiguity intrinsic to a ChRM acquired in times of mixed polarity. Consequently, the following interpretations of Alexander terrane paleogeography depend on the reference paleomagnetic poles chosen and permit Northern and Southern Hemisphere options.

### Paleolatitudes of the Alexander Terrane Relative to Laurentia, Baltica, and Siberia

Reference paleomagnetic poles of Late Silurian though Middle Devonian age for North America are located near either established Middle Silurian and Early Carboniferous paleopole positions or considerably to the west (Van der Voo, 1990; Mac Niocaill and Smethurst, 1994). The existence of a short-lived, Early Devonian loop in the apparent polar wander (APW) path is not entirely certain, but mean-European paleopoles rotated into North American coordinates appear to validate this loop. Although the loop is our favored interpretation, we consider both options in our analysis.

1. If we assume an Early Devonian loop in the North American APW path and use the mean of the poles from the western part of the loop as the Early Devonian reference paleopole for North America, this results in the loop paleolatitudes shown in Figure 3A. The paleolatitude of the Alexander terrane in the Early Devonian is either near the same latitude as it is now relative to North America (Southern Hemisphere option), as well as Baltica, or it was near the latitude of northeastern Siberia, north of all latitudes of North America (Northern Hemisphere option).

2. If we assume an Early Devonian paleopole near Middle Silurian and Early Carboniferous paleopoles, this results in the no-loop paleolatitudes shown in Figure 3A. The paleolatitude of the Alexander terrane in the Early Devonian is at the same latitude as either northwestern North America (Northern Hemisphere option), as well as Baltica, or the southwestern United States (Southern Hemisphere option).

### Paleolatitudes of the Alexander Terrane Relative to Gondwana

The Late Silurian–Early Devonian APW path for Gondwana is a matter of debate (Van der Voo, 1993). Although we favor a loop option, we consider both an interpretation with a mid-Paleozoic loop in the APW path and an interpretation assuming that the Late Silurian and Early Devonian poles are near north-central Africa.

1. If we assume an Early Devonian paleopole near the southern tip of South America, this results in the loop paleolatitudes shown in Figure 3B. The paleolatitude of the Alexander terrane is similar to the paleolatitude of eastern Australia, India, or northern Africa (Southern Hemisphere option) or entirely north of Gondwana (Northern Hemisphere option).

2. If we assume an Early Devonian paleopole near north-central Africa, this results in the no-loop paleolatitudes shown in Figure 3B. The paleolatitude of the Alexander terrane is similar to either eastern Australia (Northern Hemisphere option) or central Antarctica (Southern Hemisphere option).

## DISCUSSION AND CONCLUSIONS

Paleomagnetic data from the Lower Devonian Karheen Formation indicate that the Alexander terrane was at 14° ± 5° (north or south) paleolatitude during Early Devonian time. Due to the uncertain nature of the Devonian part of the North American and Gondwana APW paths, several positions of the Alexander terrane relative to North America (Laurentia) or Gondwana are permissible (Fig. 3). Our preferred interpretation involves a loop in both the Laurentian and Gondwana APW paths; therefore, we prefer the paleogeographies implied by comparing the Karheen paleopole to those reference paleopoles (loop paleolatitude bands in Fig. 3).

Furthermore, 1.0–3.0 Ga U-Pb ages for detrital zircons obtained from the Karheen

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Formation suggest a nearby continental source during Early Devonian time. Most of the Precambrian grains are 1.0–1.2 Ga, with a spread of ages that matches the age of igneous rocks in Grenville provinces. There is also a considerable number of grains of 1.48–1.53, 1.67–1.68, and 1.73–1.77 Ga ages. The 1.62–1.77 Ga grains also could have come from southwestern North America; however, the 1.48–1.53 Ga grains have no recognized source in North America.

The Scandinavian part of Baltica appears to be the only continent in the Northern Hemisphere during the Early Devonian with crust of the appropriate age to supply the 1.0–1.2 and 1.48–1.53 Ga zircon grains (Fig. 3A). Eastern Australia is the only Gondwana continent (Southern Hemisphere) capable of supplying the zircon grains (Fig. 3B). Both Siberia and northwestern North America lack crust of the appropriate age. An Early Devonian Alexander terrane position near the southwestern United States (Fig. 3B) appears to be a viable paleoposition that is consistent with the detrital zircon data (with the exception of the 1.48–1.53 Ga grains), but the southwestern United States is inconsistent with our preferred interpretation of the North American APW path (Fig. 3A) and it is not supported by fossil evidence (Soja, 1994).

The similarity of Silurian stromatolite-sphinctozoan reefs of the Alexander terrane, the Nixon Fork terrane, and the Ural Mountains suggests a shared Early Devonian seaway for these regions (Soja, 1994; Rigby et al., 1994). Furthermore, Savage (1990) described Paleozoic faunal affinities of the Alexander terrane suggesting that the terrane was associated with North America or northeastern Siberia throughout Silurian-Devonian time. The paleontological data rule out the possibility that the Alexander terrane was near eastern Australia during the Early Devonian. If eastern Australia and western North America are excluded, the remaining near-continent paleolatitude possibilities include a Late Silurian–Early Devonian Alexander position north (~15°) of central Alaska (Northern Hemisphere, loop option, Fig. 3A), or a position near the Scandinavian part of Baltica (Southern Hemisphere, loop option, Fig. 3A). A Baltica association is consistent with the paleomagnetic and paleontologic data and suggests that the Klakas orogeny of the Alexander terrane may have been related to the Caledonide–pre-Ellesmerian tectonic events in Arctic Canada. Furthermore, the Early Devonian Karheen red beds may have a genetic link to the "Old Red Sandstone" of present-day northern Europe. However, platform-margin microbial deposits containing the sphinctozoan sponges described by Rigby et al. (1994) are not known in the Canadian Arctic, where a Uralian seaway should have enabled transmigration between Laurentia and Baltica.

The Baltica option appears to be the only Early Devonian paleogeographic setting that satisfies the existing paleomagnetic, paleontologic, and detrital zircon data. Unfortunately, few Paleozoic plate-motion data are available to evaluate the feasibility of moving the Alexander terrane from Baltica to its current North American position, or to satisfy the additional requirement of moving the Alexander terrane to ~12° south of its present position relative to North America by Triassic time (as proposed by Haeussler et al., 1992). However, little of central Alaska was in its current position in the Paleozoic and ~150 m.y. separate the estimated Devonian and Triassic paleolatitudes. Thus, space and time were available for moving the Alexander terrane to the Triassic position. Further study of Baltic detrital zircon ages, Arctic paleontology, and the paleomagnetism of other Lower Devonian red beds may provide useful tests of this hypothesis. In addition, the Karheen Formation accumulated during the waning stages of a major orogenic event (Klakas). Thus, if Klakas deformation can be linked with Caledonide (or other) orogenesis, it may provide an additional constraint for the Early Devonian paleogeography of the Alexander terrane.

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